Exno:1b)	Version Control & Branch Management With GitHub
18.07.2025	

Aim:

To apply the fundamental concepts of Git version control and GitHub collaboration by creating, managing, and merging branches, commits, and pull requests using a **Quiz App** web application. The project demonstrates a collaborative workflow by developing the HTML structure and CSS styling in separate branches using two GitHub accounts.

Tools Required

- Git & GitHub
- Git Bash / Terminal
- Visual Studio Code
- Web Browser

Procedure

1. Create a new repository on GitHub named **Quizz-App**.
2. Clone the repository into two separate local folders to simulate two developers:

git clone https://github.com/yourusername/Quizz-App.git

Developer 1:

Quizz-App-Dev1

Developer 2:

Quizz-App-Dev2

3. In **Developer 1**, create and switch to a new branch named **html-feature**:

git checkout -b html-feature

4. Create the **index.html** file containing the basic structure of the Quiz App. Add, commit, and push the changes:

```
git add index.html
git commit -m "Add Quiz App HTML structure"
git push origin html-feature
```

5. On GitHub, create a Pull Request from **html-feature** to **main**, review it, and merge it.
6. In **Developer 2**, update the local repository if required and create a new branch named **css-feature**:

git checkout -b css-feature

7. Create the **style.css** file containing the styling for the Quiz App interface. Add, commit, and push the changes:

```
git add style.css
git commit -m "Add Quiz App CSS styling"
git push origin css-feature
```

8. Create a Pull Request from **css-feature** to **main** on GitHub.
9. If merge conflicts occur, resolve the conflicts locally, commit the resolved changes, and push the updated branch.
10. Merge the **css-feature** Pull Request into the **main** branch.
11. Verify that the **main** branch contains the following project files:

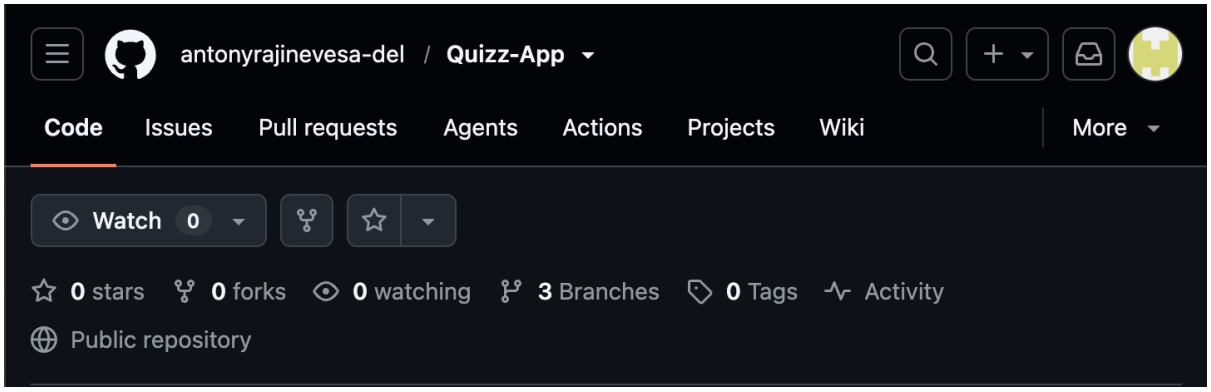
README.md
index.html
style.css

12.

1. Verify the successful completion of the project by checking:
2. Branch history
3. Commit history
4. Pull Requests
5. Merged branches on GitHub

OUTPUT:

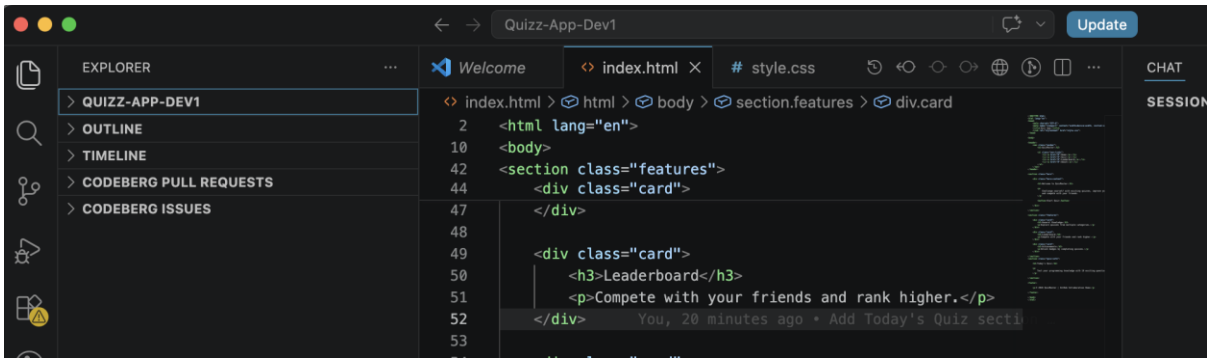
1. CREATE A REPOSITORY IN GITHUB (REMOTE) – QUIZZ-APP



2. QUIZ APP PROJECT: CLONING THE REPOSITORY

```
nevesaaxd@Nevesas-MacBook-Air Desktop % git clone https://github.com/antonyrajinevesa-del/Quizz-App.git Quizz-App-Dev1
Cloning into 'Quizz-App-Dev1'...
warning: You appear to have cloned an empty repository.
nevesaaxd@Nevesas-MacBook-Air Desktop % cd Quizz-App-Dev1
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % ls
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git config user.name "antonyrajinevesa-del"
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git config user.email "antonyrajinevesa@gmail.com"
```

3. QUIZ APP PROJECT IN VISUAL STUDIO CODE

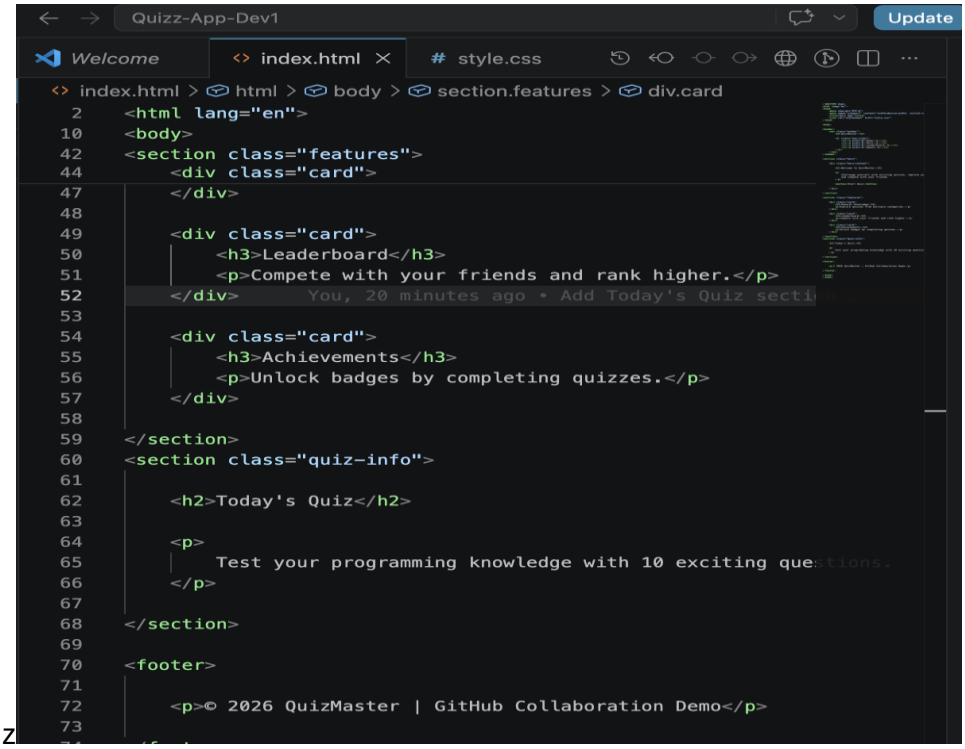


4. CREATING HTML FEATURE BRANCH

```
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git checkout -b html-feature
Switched to a new branch 'html-feature'
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git branch
* html-feature
  main
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git status
On branch html-feature
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
```

5. HTML DEVELOPMENT IN VISUAL STUDIO CODE



6.GIT COMMIT PROCESS FOR HTML FEATURE BRANCH

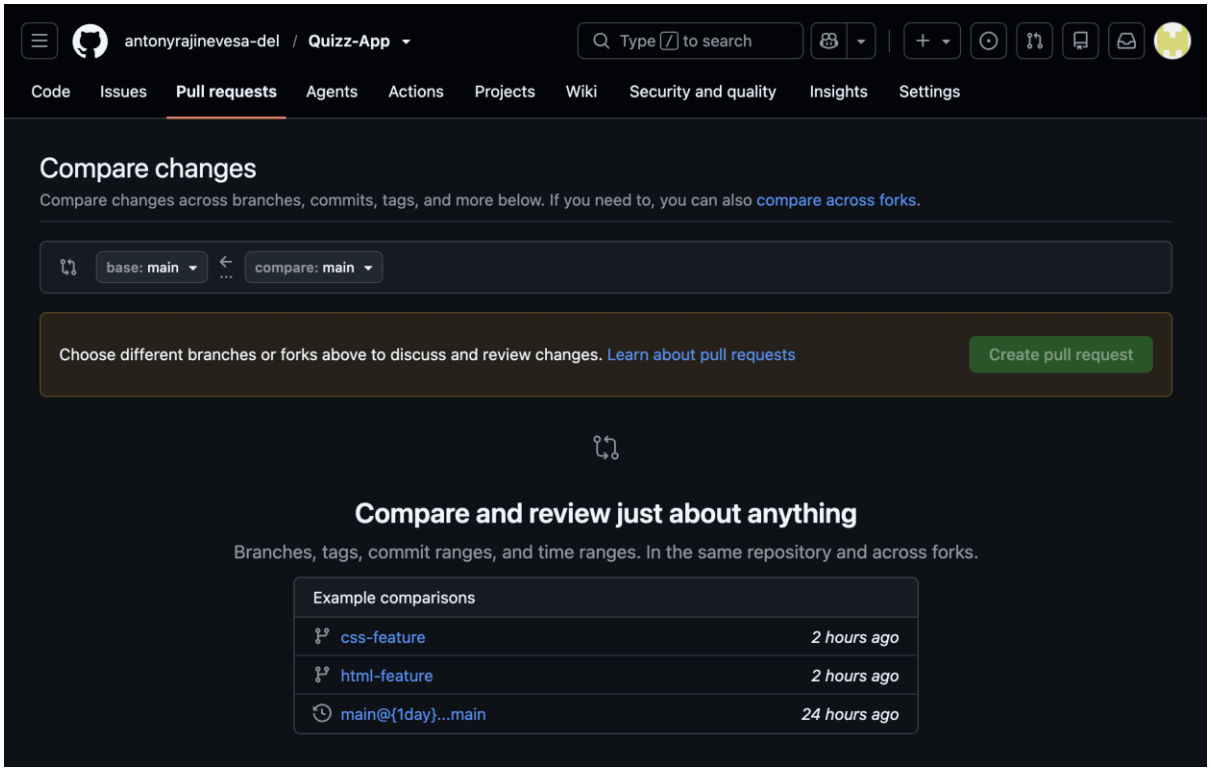
```
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git add index.html
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git commit -m "Add Today's Quiz section"
[html-feature 3becbe9] Add Today's Quiz section
 1 file changed, 77 insertions(+)
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git push -u origin html-feature
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 925 bytes | 925.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'html-feature' on GitHub by visiting:
remote:      https://github.com/antonyrajinevesa-del/Quizz-App/pull/new/html-feature
remote:
To https://github.com/antonyrajinevesa-del/Quizz-App.git
 * [new branch]      html-feature -> html-feature
branch 'html-feature' set up to track 'origin/html-feature'.
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git status
On branch html-feature
Your branch is up to date with 'origin/html-feature'.

nothing to commit, working tree clean
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % ls -l
total 8
-rw-r--r--@ 1 nevesaaxd  staff  1496 Jul 21 23:24 index.html
-rw-r--r--@ 1 nevesaaxd  staff     0 Jul 21 22:56 style.css
```

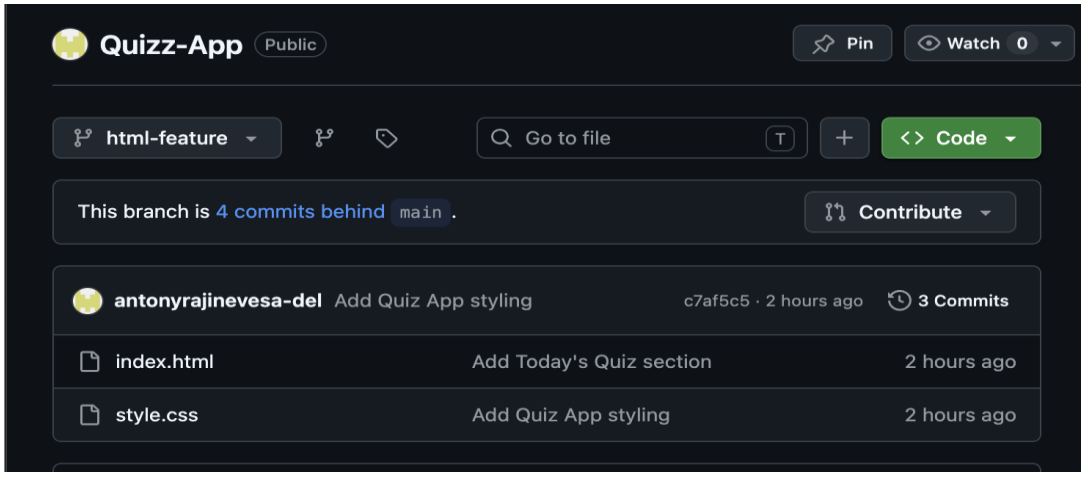
7. GIT PUSH ORIGIN HTML-FEATURE COMMAND

```
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git push -u origin html-feature
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 925 bytes | 925.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'html-feature' on GitHub by visiting:
remote:      https://github.com/antonyrajinevesa-del/Quizz-App/pull/new/html-feature
remote:
To https://github.com/antonyrajinevesa-del/Quizz-App.git
 * [new branch]      html-feature -> html-feature
branch 'html-feature' set up to track 'origin/html-feature'.
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git status
On branch html-feature
Your branch is up to date with 'origin/html-feature'.
```

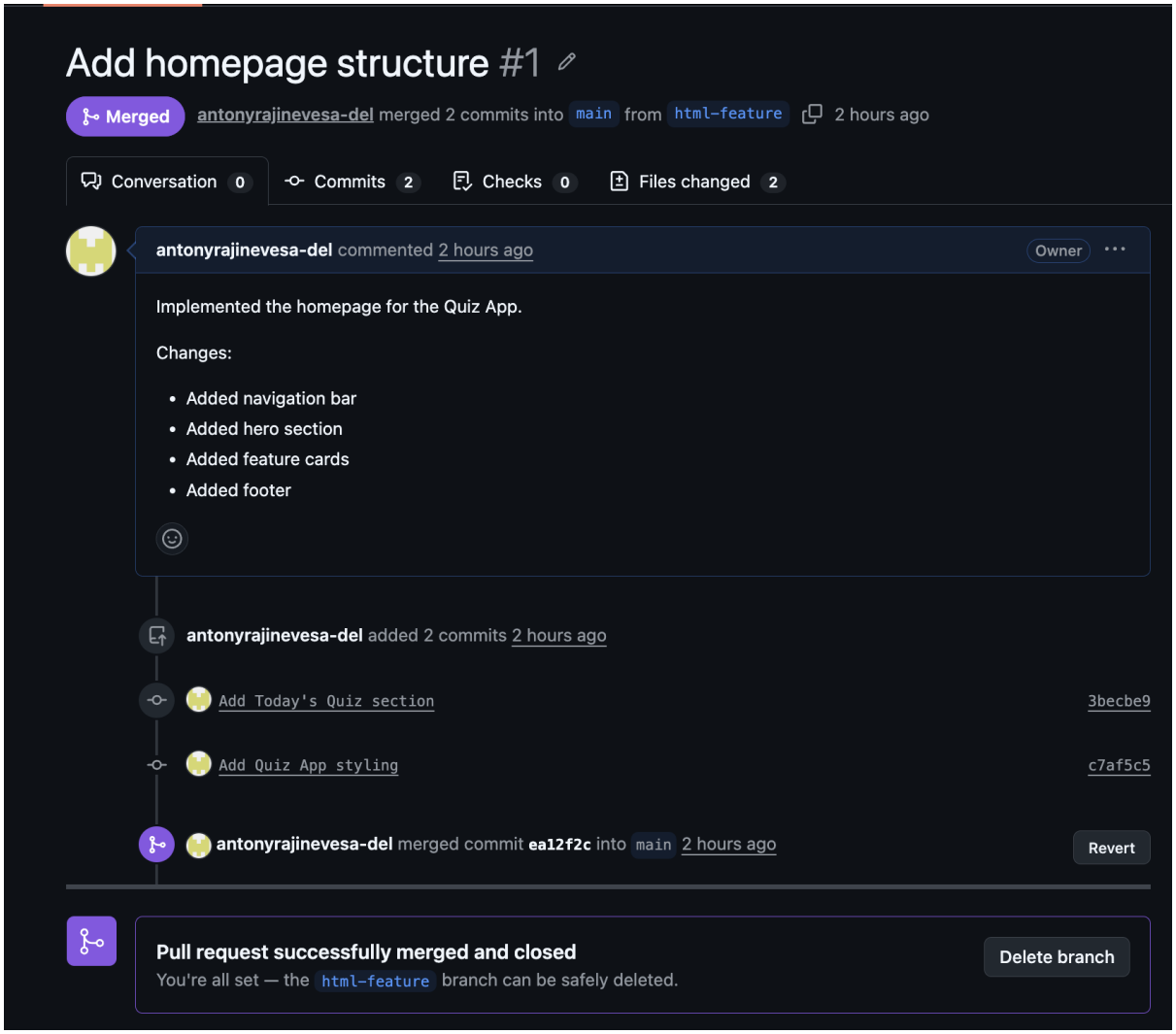
8. GITHUB PULL REQUEST PAGE FOR HTML FEATURE



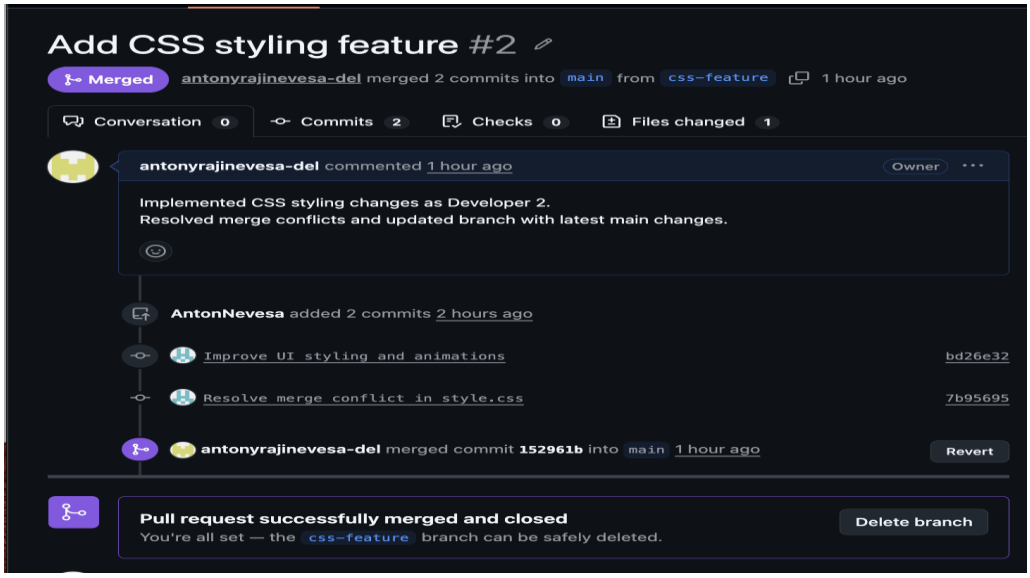
9. HTML FEATURE PULL REQUEST DETAILS



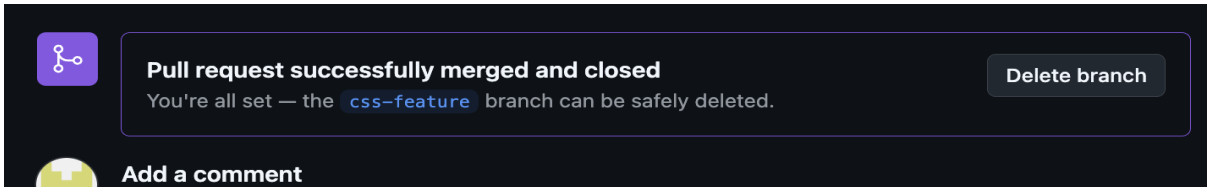
10. MERGING HTML FEATURE PULL REQUEST



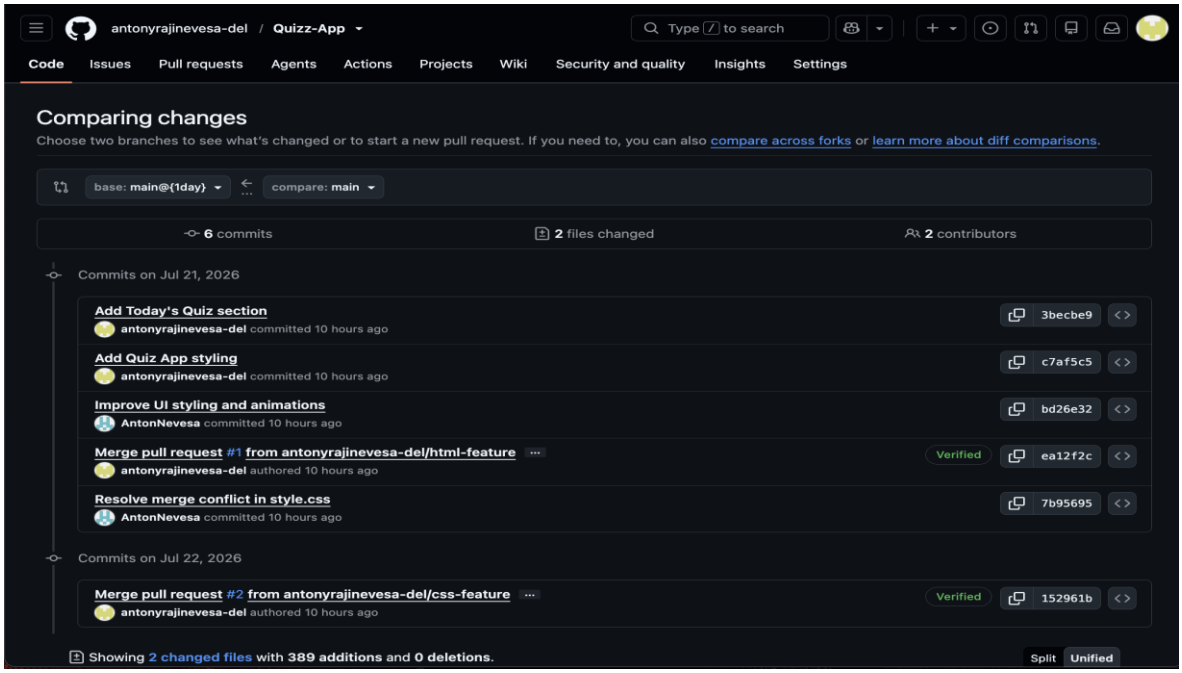
11. CSS FEATURE MERGED SUCCESSFULLY



12. MERGING CSS FEATURE



13.Comparision of Branche



14.Branch 1(Developer1-antonyrajinevesa-del)

```
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git status
On branch html-feature
Your branch is up to date with 'origin/html-feature'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   style.css

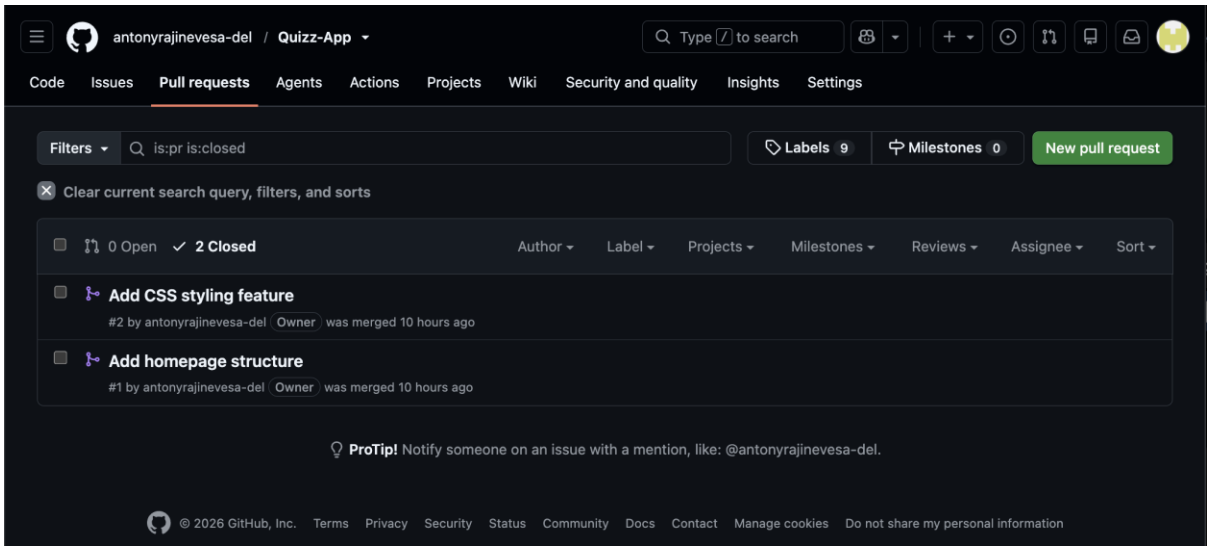
no changes added to commit (use "git add" and/or "git commit -a")
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git add style.css
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git commit -m "Add Quiz App styling"
[html-feature c7af5c5] Add Quiz App styling
 1 file changed, 164 insertions(+)
nevesaaxd@Nevesas-MacBook-Air Quizz-App-Dev1 % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 10 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 879 bytes | 879.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/antonyrajinevesa-del/Quizz-App.git
   3becbe9..c7af5c5  html-feature -> html-feature
```

15.User Interface





16. PULL REQUEST HISTORY



RUBRICS:

Github commands syntax & description (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

RESULT:

Thus, Git and GitHub commands were successfully practiced to implement version

control, manage branches, and merge pull requests through a Daily Task Tracker Application using a JavaScript backend.

EX.NO:2	DEVELOPING A RESTFUL CRUD API USING NODE.JS, EXPRESS.JS, MONGODB ATLAS, AND MONGOOSE
DATE:	

Aim

To develop a Backend API using **Node.js**, **Express.js**, **MongoDB Atlas**, **Mongoose**, **GitHub**, and **Thunder Client** for performing CRUD operations.

Step 1: Install Required Software

Install the following software.

- Node.js (LTS Version)
- Visual Studio Code
- Git
- MongoDB Atlas Account
- Thunder Client Extension in VS Code

Verify Node.js installation.

node -v

npm -v

Verify Git installation.

git --version

Step 2: Create MongoDB Atlas Cluster

1. Login to MongoDB Atlas.
2. Create a Free Cluster (M0).
3. Create Database User.

Example

Username : nevesa

Password : nevesa2026

4. Go to

Security

↓

Network Access

Add

0.0.0.0/0

5. Click

Connect

↓

Drivers

Copy the connection string.

Example

mongodb+srv://rajkumar:nevesa2026@cluster0.yekjugz.mongodb.net/labdb?retryWrites=true&w=majority&appName=Cluster0

Step 3: Create Node.js Project

Open Command Prompt.

Create Project Folder.

mkdir api

Move to project folder.

cd api

code .

VScode Terminal

Initialize Node Project.

npm init -y

Initialize Git.

git init

Step 4: Install Packages

Install Express.

npm install express

Install Mongoose.

npm install mongoose

Install CORS.

npm install cors

Install Body Parser.

npm install body-parser

Install Nodemon.

npm install --save-dev nodemon

Step 6: Create Folder Structure

API

```
│
├── node_modules
├── models
│     └── Item.js
│
├── routes
│     └── itemRoutes.js
│
├── server.js
├── package.json
└── package-lock.json
```

Step 7: Modify package.json

Replace the scripts section with

```
"scripts": {
   "start": "nodemon server.js"
},
```

Step 8: Create Item Model

Create

models → Item.js

item.js

```
const mongoose = require('mongoose');
const ItemSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true
  },
  quantity: {
    type: Number,
    required: true
  }
});
module.exports = mongoose.model('Item', ItemSchema);
```

Step 9: Create Routes

Create

routes → itemRoutes.js

```
const express = require('express');
const router = express.Router();
const Item = require('../models/Item');
router.post('/', async (req, res) => {
  try {
    const newItem = new Item(req.body);
    await newItem.save();
    res.status(201).json(newItem);
  } catch (err) {
    res.status(400).json({ message: err.message });
  }
```

```
});
router.get('/', async (req, res) => {
  try {
    const items = await Item.find();
    res.json(items);
  } catch (err) {
    res.status(500).json({ message: err.message });
  }
});
router.put('/:id', async (req, res) => {
  try {
    const updatedItem = await Item.findByIdAndUpdate(
      req.params.id,
      req.body,
      { new: true }
    );
    res.json(updatedItem);
  } catch (err) {
    res.status(400).json({ message: err.message });
  }
});
router.delete('/:id', async (req, res) => {
  try {
    await Item.findByIdAndDelete(req.params.id);
    res.json({ message: "Item deleted" });
  } catch (err) {
    res.status(500).json({ message: err.message });
```

```
    }
});
module.exports = router;
```

Step 10: Create Server

Create

server.js

```
const express = require('express');
const mongoose = require('mongoose');
const bodyParser = require('body-parser');
const cors = require('cors');
const app = express();

app.use(cors());
app.use(bodyParser.json());
mongoose.connect(
"mongodb+srv://rajkumar:Rajkumar2026@cluster0.yekjugz.mongodb.net/labdb?retryWrites=true&w=majority&appName=Cluster0"
)
.then(() => console.log("MongoDB Atlas Connected"))
.catch(err => console.log(err));
const itemRoutes = require('./routes/itemRoutes');
app.use('/api/items', itemRoutes);
app.listen(5000, () => {
    console.log("Server running on port 5000");
});
```

Step 11: Start Server

Run

npm start

Output

Server running on port 5000

MongoDB Atlas Connected

Step 12: Test API using Thunder Client

POST

Method

POST

URL

http://localhost:5000/api/items

Body

```
{
   "name":"Book",
   "quantity":10
}
```

Click **Send**.

Response

```
{
   "_id":"6a57278f4a3de59dfccd8939",
   "name":"Book",
   "quantity":10,
   "__v":0
}
```

GET

Method

GET

URL

http://localhost:5000/api/items

Click **Send**.

Response


```
[
  {
    "_id":"6a57278f4a3de59dfccd8939",
    "name":"Book",
    "quantity":10,
    "__v":0
  }
]
```


PUT

Method

PUT

URL

http://localhost:5000/api/items/6a57278f4a3de59dfccd8939

Body


```
{
  "name":"Book",
  "quantity":20
}
```


Click **Send**.

Response


```
{
  "_id":"6a57278f4a3de59dfccd8939",
  "name":"Book",
  "quantity":20,
```

"__v":0

}

DELETE

Method

DELETE

URL

http://localhost:5000/api/items/6a57278f4a3de59dfccd8939

Click **Send**.

Response

{

"message":"Item deleted"

}

Step 13: Verify in MongoDB Atlas

Open

Database

↓

Browse Collections

↓

labdb

↓

items

Verify

- POST → New document inserted.
- GET → Documents retrieved.
- PUT → Quantity updated.
- DELETE → Document deleted.

Step 14: Push Project to GitHub

Check status.

git status

Add files.

git add .

Commit.

git commit -m "Completed Backend API CRUD"

Rename branch.

git branch -M main

Add remote.

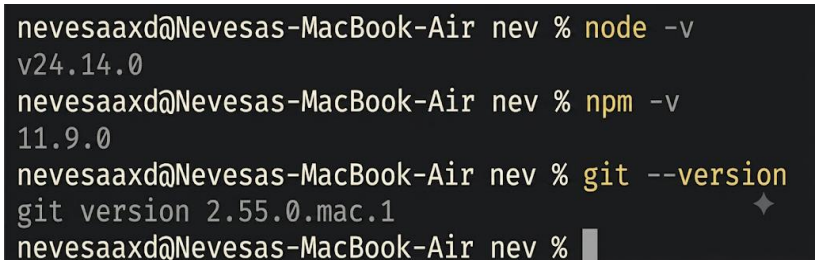
git remote add origin https://github.com/<username>/<repository>.git

Push.

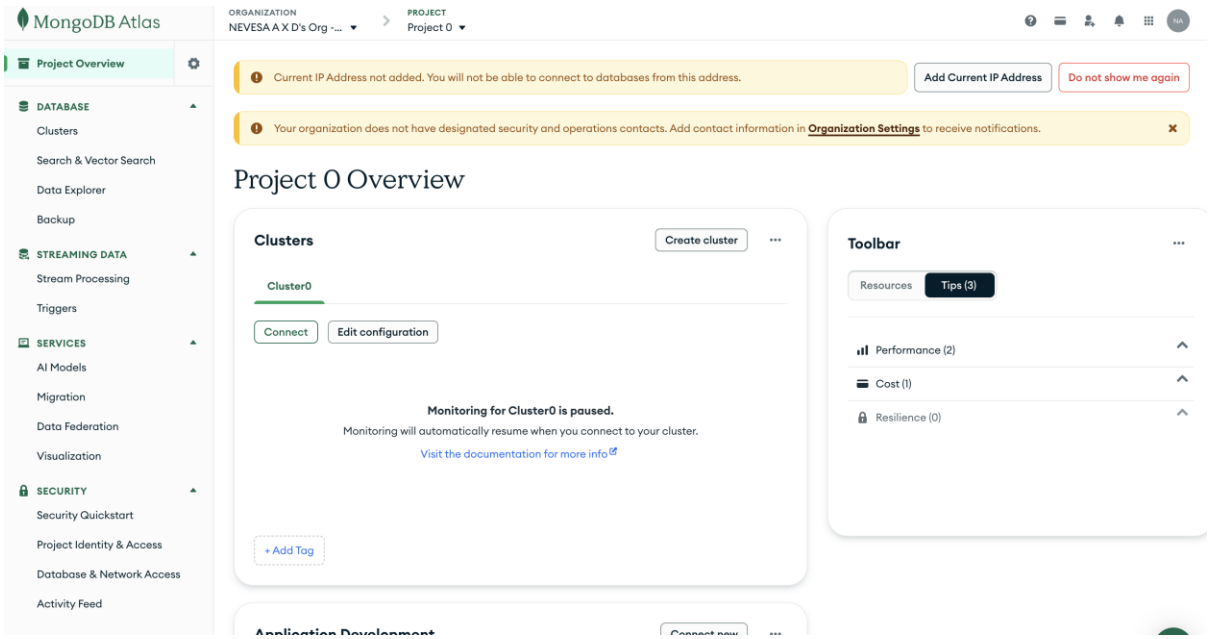
git push -u origin main

OUTPUT:

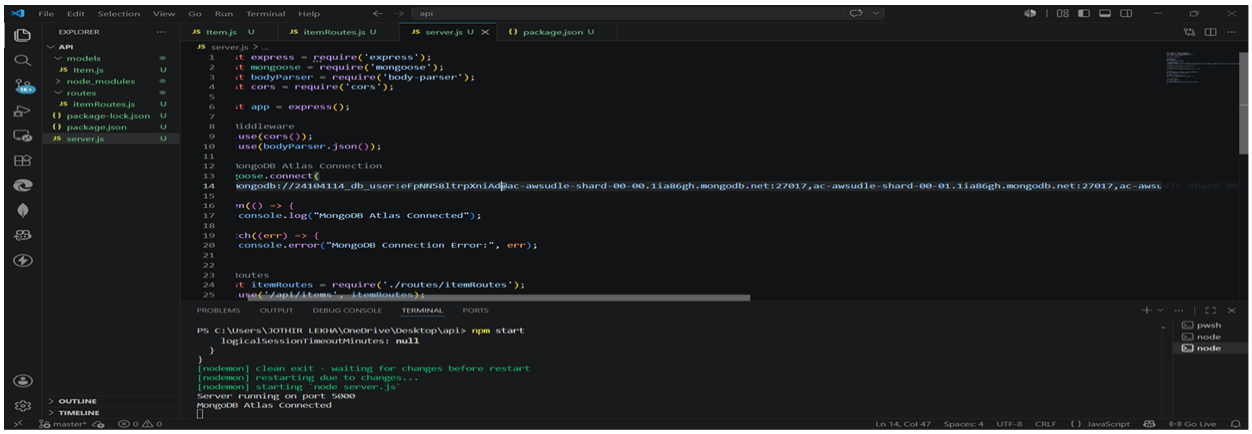
VERIFY NODE.JS AND GIT INSTALLATION:



MONGODB ATLAS CLUSTER:



PROJECT FOLDER IN VS CODE:



INSTALLED PACKAGES:

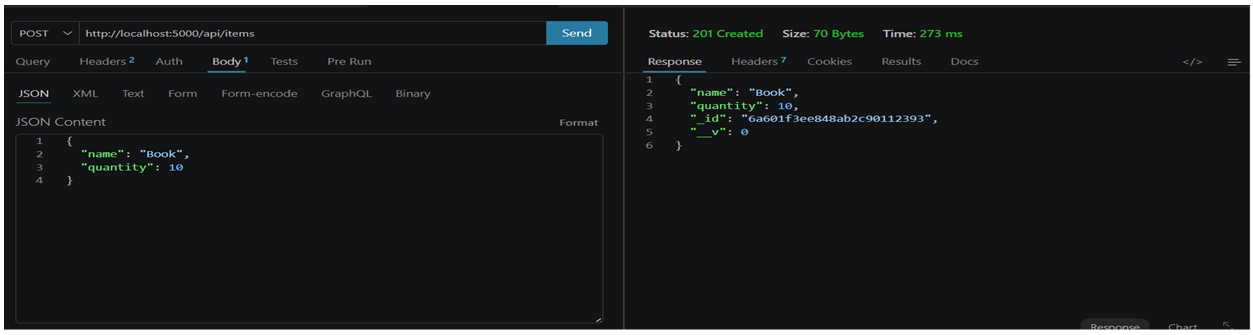
```
nevesaaxd@Nevesas-MacBook-Air nev % npm install mongodb
up to date, audited 114 packages in 3s

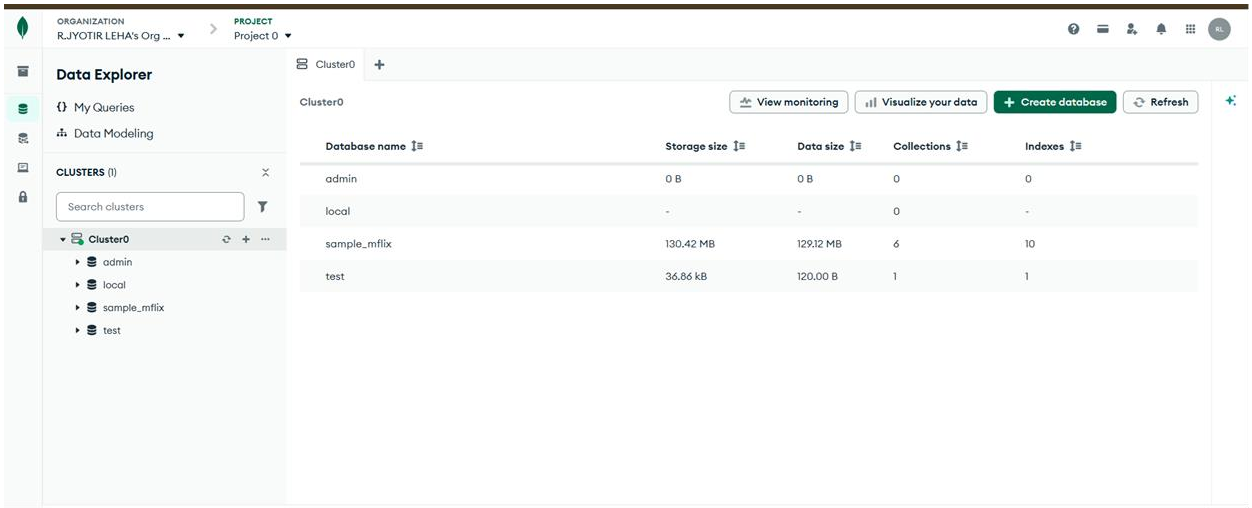
32 packages are looking for funding
  run `npm fund` for details
found \e[32;1m0\e[0m vulnerabilities
```

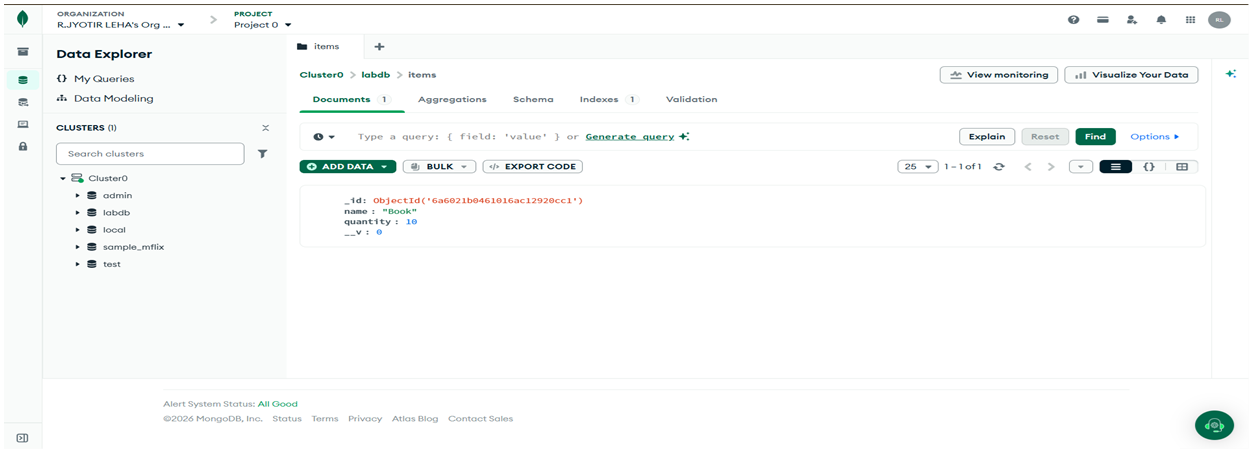
RUNNING SERVER:

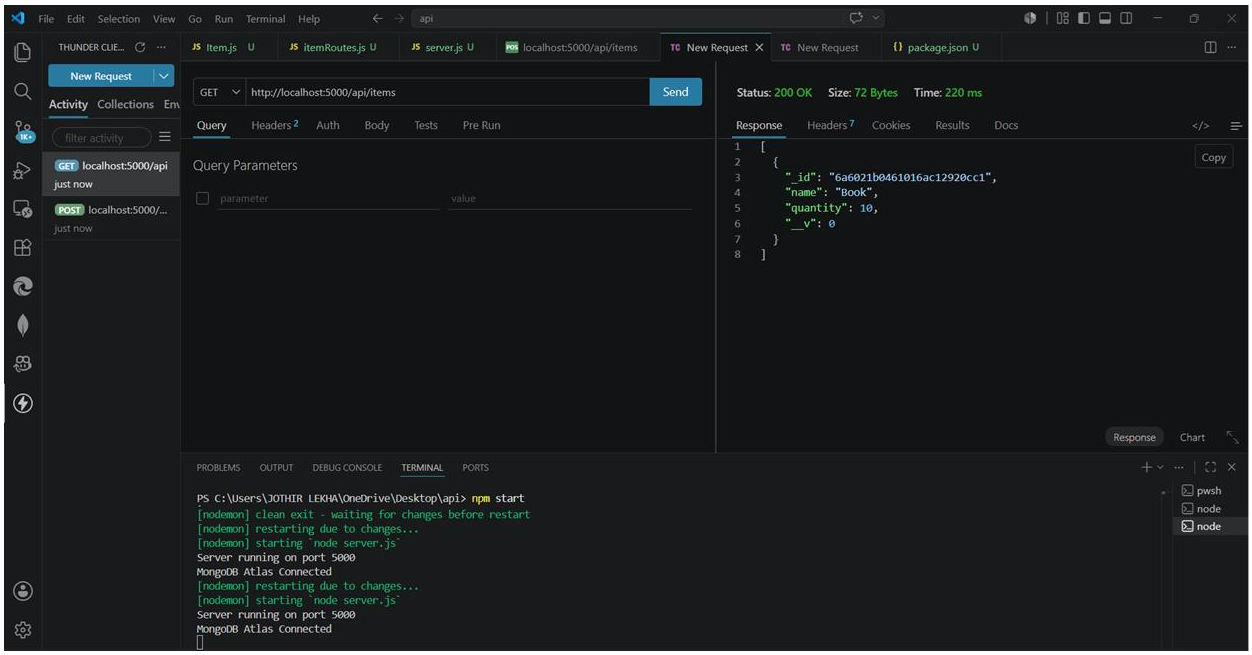
```
[nodemon] clean exit - waiting for changes before restart
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Atlas Connected
```

REQUEST COMMANDS:

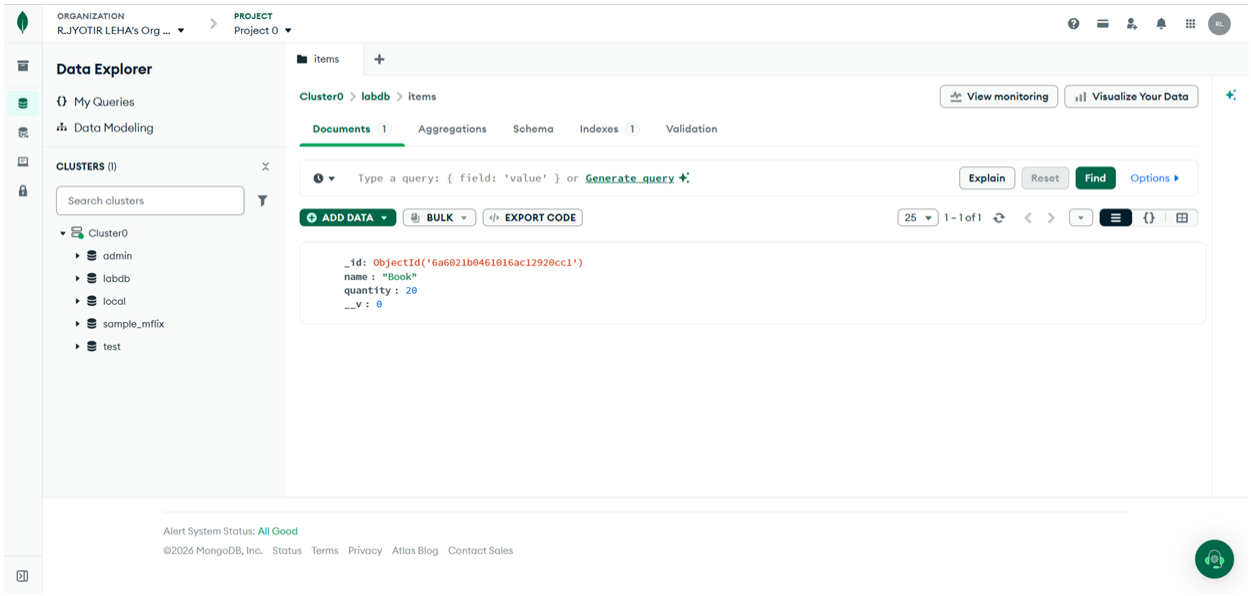




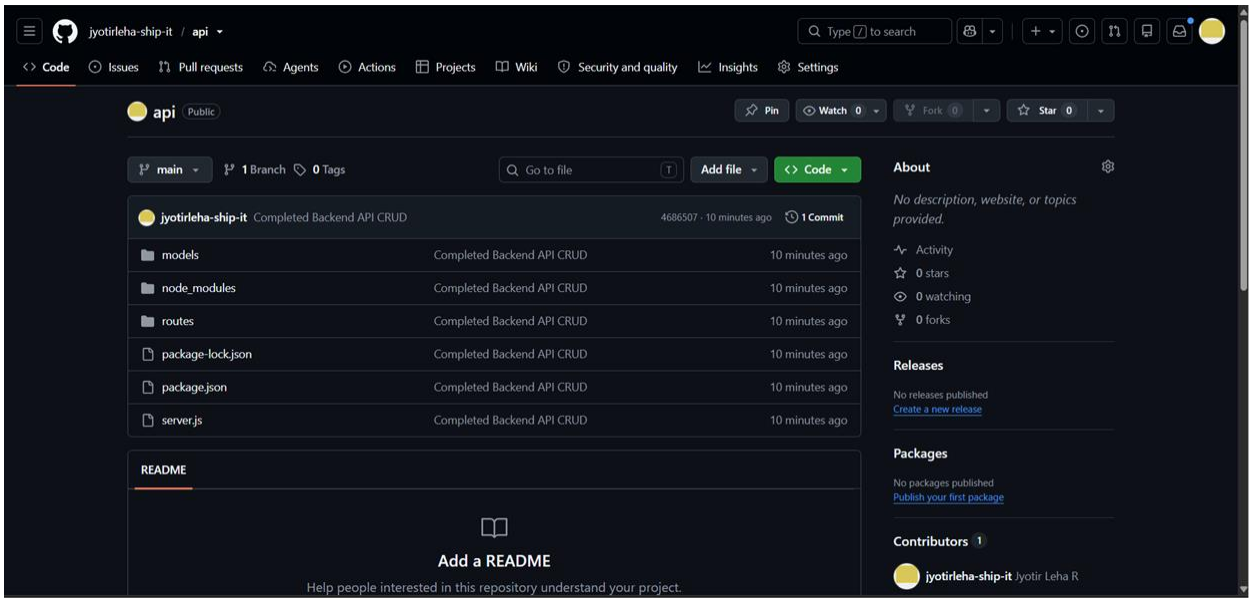




MONGODB ATLAS VERIFICATION:


ORGANIZATION
R.JYOTIR LEHA's Org ...
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
admin
labdb
local
sample_mflix
test
items
Cluster0 > labdb > items
View monitoring
Visualize Your Data
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('6a6021b0461016ac12920cc1')
name : "Book"
quantity : 20
__v : 0
Alert System Status: All Good
©2026 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales

GITHUB PUSH:


jyotirleha-ship-it / api
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
api
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
jyotirleha-ship-it Completed Backend API CRUD
4686507 · 10 minutes ago
1 Commit
models Completed Backend API CRUD 10 minutes ago
node_modules Completed Backend API CRUD 10 minutes ago
routes Completed Backend API CRUD 10 minutes ago
package-lock.json Completed Backend API CRUD 10 minutes ago
package.json Completed Backend API CRUD 10 minutes ago
server.js Completed Backend API CRUD 10 minutes ago
README
Add a README
Help people interested in this repository understand your project.
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
jyotirleha-ship-it Jyotir Leha R

RUBRICS:

Github commands syntax & description (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)